

Theorem (Maximum-norm a posteriori bound at final time via elliptic indicators). Let u be the solution of $\partial_t u + \mathcal{M}u = F$ on $\Omega \times (0, T]$ with $u|_{\partial\Omega} = 0$ and $u(\cdot, 0) = u_0$, and let $\{u_h^j\}_{j=0}^M \subset V_h^0$ be the backward Euler/FEM solution with time steps $0 = t_0 < t_1 < \dots < t_M = T$. Let $u_h \in W_2^1(0, T; H_0^1(\Omega), L^2(\Omega))$ be any admissible extension of the discrete values, meaning $u_h(\cdot, t_j) = u_h^j$ for all j and u_h is regular enough so that the error $e := u - u_h$ satisfies the hypothesis of Lemma ?? and admits point-wise evaluation in x at time T . Define the error quantity at time T by $e(\cdot, T) = u(\cdot, T) - u_h(\cdot, T)$. Then there exists a constant C depending only on the data of the problem (in particular the operator $\mathcal{K}$ and the Green-function bounds in Assumption ??) such that

$$\|u(T) - u_h(T)\|_{\infty, \Omega} \leq C \left(\|u_0 - u_h(\cdot, 0)\|_{\infty, \Omega} + \sum_{j=1}^M (\eta_{\text{ell}}^j A_j + \eta_{\text{ell}, \delta_t}^j B_j) \right),$$

where $I_j = (t_{j-1}, t_j)$ and

$$A_j := \int_{I_j} \|\Gamma(T - s)\|_{1, \Omega} ds, \quad B_j := \int_{I_j} \|\partial_t \Gamma(T - s)\|_{1, \Omega} ds,$$

with $\eta_{\text{ell}}^j = \eta(u_h^j, f^j + \psi_h^j)$ and $\eta_{\text{ell}, \delta_t}^j$ the estimator for $\delta_t u_h^j$ (equivalently the stated maximum-norm bound for $\delta_t(R^j - u_h^j)$).

Proof. Set $e(t) := u(t) - u_h(t)$. By the admissibility of u_h , we have $e \in W_2^1(0, T; H_0^1(\Omega), L^2(\Omega))$, so Lemma ?? applies to $w = e$. For each $t \in (0, T]$, Lemma ?? yields, for almost every $x \in \Omega$,

$$e(x, t) = (\Gamma(t), e(0)) + \int_0^t \langle \mathcal{K}e(s), \Gamma(t - s) \rangle ds.$$

Taking $t = T$ gives

$$e(x, T) = (\Gamma(T), e(0)) + \int_0^T \langle \mathcal{K}e(s), \Gamma(T - s) \rangle ds.$$

Taking the L^∞ -norm in x and using the triangle inequality gives

$$\|e(T)\|_{\infty, \Omega} \leq |(\Gamma(T), e(0))| + \int_0^T |\langle \mathcal{K}e(s), \Gamma(T - s) \rangle| ds.$$

We estimate the initial term using the Green-function bound in Assumption ?. Since $\Gamma(T) \in L^1(\Omega)$ with $\|\Gamma(T)\|_{1, \Omega} \leq \kappa_0 e^{-\gamma T} = \phi_{0, \Gamma}(T)$, we have

$$|(\Gamma(T), e(0))| \leq \|\Gamma(T)\|_{1, \Omega} \|e(0)\|_{\infty, \Omega} \leq \phi_{0, \Gamma}(T) \|u_0 - u_h(\cdot, 0)\|_{\infty, \Omega}.$$

It remains to bound the convolution term. To connect $\langle \mathcal{K}e(s), \Gamma(T-s) \rangle$ to elliptic reconstructions and time-discretization defects, we use the weak formulations. By definition, u satisfies

$$\frac{d}{dt}(u(t), v) + c(u(t), v) = \langle F(t), v \rangle \quad \text{for all } v \in V \text{ and a.e. } t,$$

while u_h satisfies the same identity only at nodes in the discrete sense. Therefore, for almost every s , the functional $\langle \mathcal{K}e(s), v \rangle$ equals the residual of the u_h insertion into the continuous weak formulation, namely

$$\langle \mathcal{K}e(s), v \rangle = \langle F(s), v \rangle - \frac{d}{dt}(u_h(s), v) - c(u_h(s), v) \quad \text{for all } v \in V.$$

Now fix an interval $I_j = (t_{j-1}, t_j)$ and use the piecewise-linear interpolation of u_h in time with respect to the discrete backward Euler structure (this is precisely the role of choosing an admissible extension consistent with the time reconstruction). For $s \in I_j$, the time derivative term contributes the discrete time defect $\delta_t(R^j - u_h^j)$ after introducing the spatial reconstruction R^j and its elliptic defect ψ_h^j . Concretely, R^j is defined by $c(R^j, v) = \langle f^j + \psi_h^j, v \rangle$, and ψ_h^j is defined by $c_h(\psi_h^j, v_h) = c_h(u_h^j, v_h) - (f^j, v_h)_h$. Using these definitions, one rewrites the spatial part of the residual in V^* as the elliptic residual acting on the reconstruction mismatch $R^j - u_h^j$. Furthermore, because the extension u_h between nodes is chosen so that its time derivative on I_j is consistent with the discrete backward Euler difference, the remaining time derivative mismatch can be transferred onto the time derivative of the reconstruction mismatch, yielding a decomposition of the residual pairing of the form

$$\langle \mathcal{K}e(s), \Gamma(T-s) \rangle = c(R^j - u_h^j, \Gamma(T-s)) + (\delta_t(R^j - u_h^j), \Gamma(T-s)) \quad \text{for a.e. } s \in I_j.$$

(Here we use the standard identification of the continuous bilinear form c with the weak action of $\mathcal{M}$, and the fact that the dual action of the time derivative term corresponds to integration by parts in time on each slab, producing a Γ -factor with a time derivative; the admissibility of u_h ensures the required differentiations are justified.) Taking absolute values and using the properties of c and the definition of the duality action, we obtain for $s \in I_j$ bounds of the type

$$|\langle \mathcal{K}e(s), \Gamma(T-s) \rangle| \leq C \left(\|R^j - u_h^j\|_{\infty, \Omega} \|\Gamma(T-s)\|_{1, \Omega} + \|\delta_t(R^j - u_h^j)\|_{\infty, \Omega} \|\partial_t \Gamma(T-s)\|_{1, \Omega} \right).$$

Integrating over $s \in I_j$ and summing over $j = 1, \dots, M$ gives

$$\int_0^T |\langle \mathcal{K}e(s), \Gamma(T-s) \rangle| ds \leq C \sum_{j=1}^M \left(\|R^j - u_h^j\|_{\infty, \Omega} A_j + \|\delta_t(R^j - u_h^j)\|_{\infty, \Omega} B_j \right).$$

Finally, apply the maximum-norm reconstruction bounds and Assumption ?? . By definition of the elliptic reconstruction and the estimator, we have

$$\|R^j - u_h^j\|_{\infty, \Omega} \leq \eta_{\text{ell}}^j$$

and by the analogous discrete-time reconstruction estimate,

$$\|\delta_t(R^j - u_h^j)\|_{\infty, \Omega} \leq \eta_{\text{ell}, \delta_t}^j.$$

Therefore

$$\int_0^T |\langle \mathcal{K}e(s), \Gamma(T-s) \rangle| ds \leq C \sum_{j=1}^M \left(\eta_{\text{ell}}^j A_j + \eta_{\text{ell}, \delta_t}^j B_j \right).$$

Inserting the estimate of the initial term and the convolution term into the inequality for $\|e(T)\|_{\infty, \Omega}$ yields

$$\|u(T) - u_h(T)\|_{\infty, \Omega} = \|e(T)\|_{\infty, \Omega} \leq C \left(\|u_0 - u_h(\cdot, 0)\|_{\infty, \Omega} + \sum_{j=1}^M \left(\eta_{\text{ell}}^j A_j + \eta_{\text{ell}, \delta_t}^j B_j \right) \right),$$

which is the desired upper bound. The constants depend only on coercivity/boundedness of c and on the Green-function norms through A_j, B_j , which are controlled by Assumption ?? via $\|\Gamma(t)\|_{1, \Omega} \lesssim \phi_{0, \Gamma}(t)$ and $\|\partial_t \Gamma(t)\|_{1, \Omega} \lesssim \phi_{1, \Gamma}(t)$.

It remains to indicate dependence on the extension choice. The bound uses only $e(0) = u_0 - u_h(\cdot, 0)$ and, on each slab, the property that u_h is extended in time so that its time derivative mismatch against the backward Euler structure can be transferred into $\delta_t(R^j - u_h^j)$ in the residual pairing; moreover, admissibility is needed so that Lemma ?? applies to $w = e$ and so that $e(T)$ has a well-defined pointwise-in- x representative. Thus the argument requires u_h to be regular enough for $e \in W_2^1(0, T; H_0^1(\Omega), L^2(\Omega))$ and compatible with the discrete-time reconstruction on each I_j so that the residual decomposition into elliptic and time-discretization defect terms holds exactly.